

TP GUIDED: Implementation of a CI/CD Pipeline with GitHub Actions

Students : 2SCL / BI

1. OBJECTIVE

The objective of this lab is to implement a simple CI/CD pipeline for a static web application using GitHub Actions. The student will learn continuous integration, continuous testing, artifact generation, and deployment using GitHub Pages.

2. PREREQUISITES

- A GitHub account
- Git installed on local machine
- Basic knowledge of HTML, CSS, JavaScript
- Basic Git commands (add, commit, push)
- A code editor such as VS Code

3. PROJECT SETUP

Step 1: Create project folder

 .github	Modifié le : 29/05/2026 10:04
 app Type : Fichier source JavaScript	Modifié le : 29/05/2026 10:17 Taille : 134 octet(s)
 index	Modifié le : 29/05/2026 09:29 Taille : 324 octet(s)
 style Type : Fichier source CSS	Modifié le : 29/05/2026 09:30 Taille : 157 octet(s)

Step 2: Create application files

index.html:

- Simple HTML page with a button and script reference

```
C: > Users > hp > Desktop > CI-CD BI-2SCL > <> index.html > 📄 html
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>My First Web App</title>
5      <link rel="stylesheet" href="style.css">
6  </head>
7  <body>
8
9      <h1>Simple CI/CD Demo</h1>
10
11     <button onclick="showMessage()">
12         Click Me
13     </button>
14
15     <p id="message"></p>
16
17     <script src="app.js"></script>
18
19 </body>
20 </html>
```

app.js:

```
 Casa.cfg ● <> index.html # style.css JS app.js ✕
C: > Users > hp > Desktop > CI-CD BI-2SCL > JS app.js > 📄 showMessage
1  function showMessage() {
2      document.getElementById("message").innerHTML =
3          "CI/CD with GitHub Actions works version 2!";
4  }
```

style.css (optional):

- Basic styling for the page

```
C: > Users > hp > Desktop > CI-CD BI-2SCL > # style.css > button
1  body {
2      font-family: Arial, sans-serif;
3      text-align: center;
4      margin-top: 100px;
5  }
6
7  button {
8      padding: 10px 20px;
9      font-size: 16px;
10 }
```

4. INITIALIZE GIT AND CONNECT TO GITHUB

git init

git add .

git commit -m "Initial commit"

git remote add origin <your-repo-url>

git branch -M main

git push -u origin main

```

hp@DESKTOP-A0NOD3R MINGW64 ~
$ cd Desktop/

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop
$ cd CI
CI-CD BI-2SCL/          CISCO_LABS/          Cisco Packet Tracer.lnk

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop
$ cd CI-CD\ BI-2SCL/

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL
$ git init
Initialized empty Git repository in C:/Users/hp/Desktop/CI-CD BI-2SCL/.git/

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git add .

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git commit -m "change1"
[master (root-commit) 43ab04c] change1
3 files changed, 34 insertions(+)
create mode 100644 app.js
create mode 100644 index.html
create mode 100644 style.css

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git remote add origin https://github.com/SouadSADKI/ci-cd-2SCL_BI.git

```

Then Push:

```

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git push -u origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 711 bytes | 237.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SouadSADKI/ci-cd-2SCL_BI.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

```

TIP : ACCESS TOKEN FOR THE REMOTE REPO

The screenshot shows the GitHub Developer Settings page. On the left sidebar, under 'Personal access tokens', the 'Fine-grained tokens' option is selected. The main content area is titled 'Fine-grained personal access tokens' and includes a 'Generate new token' button. Below this, a message states: 'These are fine-grained, repository-scoped tokens suitable for personal API use and for using Git over HTTPS.' A single token is displayed with the name 'accesscid_2SCL_BI', a red key icon, and a 'Delete' button. The token's details indicate it was 'Last used within the last week' and 'Expires on Sun, Jun 28 2026'. The footer of the page shows the GitHub logo, copyright notice '© 2026 GitHub, Inc.', and links for Terms, Privacy, Security, Status, Community, Docs, Contact, Manage cookies, and Do not share my personal information.

Access on  [SouadSADKI](#)

Edit

Repository access

SouadSADKI/ci-cd-2SCL_BI

User permissions

This token does not have any user permissions.

Repository permissions

✓ Read access to metadata

✓ Read and Write access to actions and code

5. CI/CD PIPELINE IMPLEMENTATION (GITHUB ACTIONS)

Create the following file:

`.github/workflows/ci-cd.yml`

ci-cd-2SCL_BI / .github / workflows / ci-cd.yml 

 SouadSADKI Update ci-cd.yml 

Code

Blame

98 lines (78 loc) · 1.95 KB



```
1   name: Simple CI/CD Pipeline
2
3   on:
4     push:
5       branches:
6         - master
7
8   jobs:
9
10    # =====
11    # 1. BUILD STAGE
12    # =====
13    build:
14      runs-on: ubuntu-latest
15
16      steps:
17        - name: Checkout code
18          uses: actions/checkout@v4
19
20        - name: Validate project structure
21          run: |
22            echo "Checking files..."
23            test -f index.html
24            test -f style.css
25            test -f app.js
26            echo "Build stage completed successfully!"
27  --
```

```
28     # =====
29     # 2. TEST STAGE
30     # =====
31     test:
32         runs-on: ubuntu-latest
33         needs: build    # runs after build
34
35         steps:
36             - name: Checkout code
37               uses: actions/checkout@v4
38
39             - name: Basic tests
40               run: |
41                 echo "Running simple tests..."
42
43                 grep -q "<html>" index.html
44                 grep -q "function" app.js
45
46                 echo "All tests passed!"
47
48     # =====
49     # 3. PACKAGE STAGE
50     # =====
51     package:
52         runs-on: ubuntu-latest
53         needs: test
54
55         steps:
56             - name: Checkout code
57               uses: actions/checkout@v4
58
59             - name: Create deployment package
```

```
60         run: |
61             mkdir build
62             cp index.html build/
63             cp style.css build/
64             cp app.js build/
65             echo "Package created"
66
67     - name: Upload artifact
68       uses: actions/upload-artifact@v4
69       with:
70         name: web-app
71         path: build/
72
73     # =====
74     # 4. DEPLOY STAGE (GitHub Pages)
75     # =====
76     deploy:
77       runs-on: ubuntu-latest
78       needs: package
79
80       permissions:
81         contents: read
82         pages: write
83         id-token: write
84
85       steps:
86         - name: Checkout code
87           uses: actions/checkout@v4
88
89         - name: Setup GitHub Pages
90           uses: actions/configure-pages@v5
91
92         - name: Upload Pages artifact
93           uses: actions/upload-pages-artifact@v3
94           with:
95             path: .
96
97         - name: Deploy to GitHub Pages
98           uses: actions/deploy-pages@v4
```

Pipeline stages:

BUILD STAGE

- Checkout code
- Verify required files exist (index.html, app.js)

TEST STAGE

- Validate basic structure of HTML and JS files using simple grep checks

PACKAGE STAGE

- Create a build folder
- Copy application files into build directory
- Upload build artifact

6. WORKFLOW EXECUTION

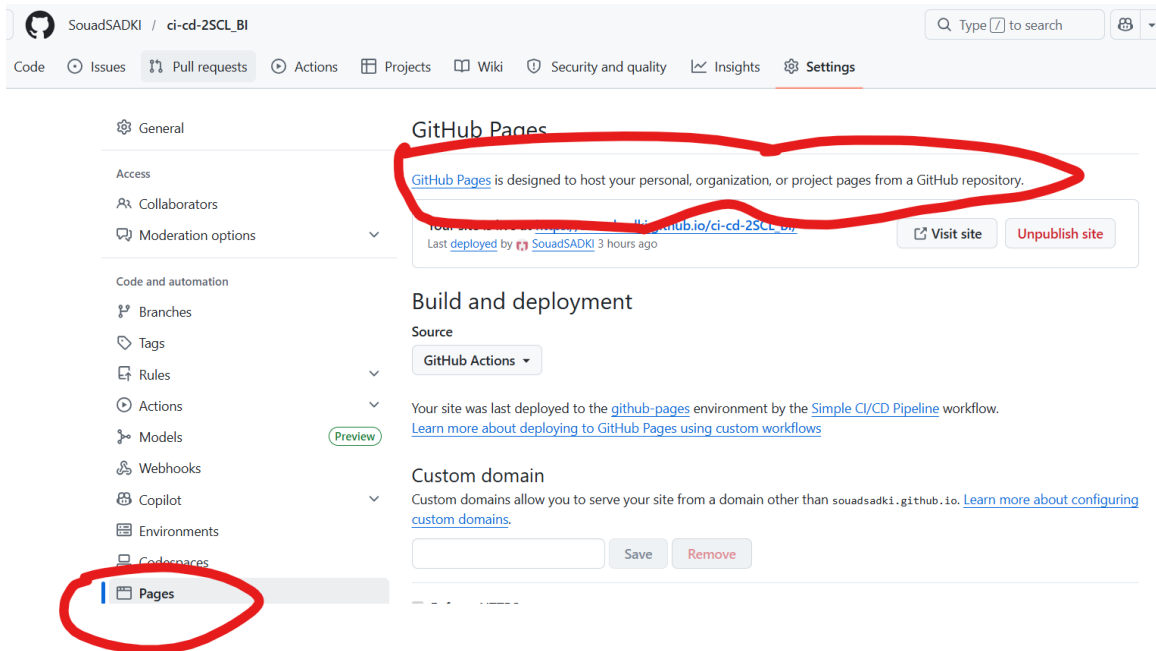
After pushing the workflow file:

- GitHub Actions automatically triggers on push to main branch
- Each job runs sequentially based on dependencies
- Build must pass before test runs
- Test must pass before packaging

7. DEPLOYMENT WITH GITHUB PAGES

Steps:

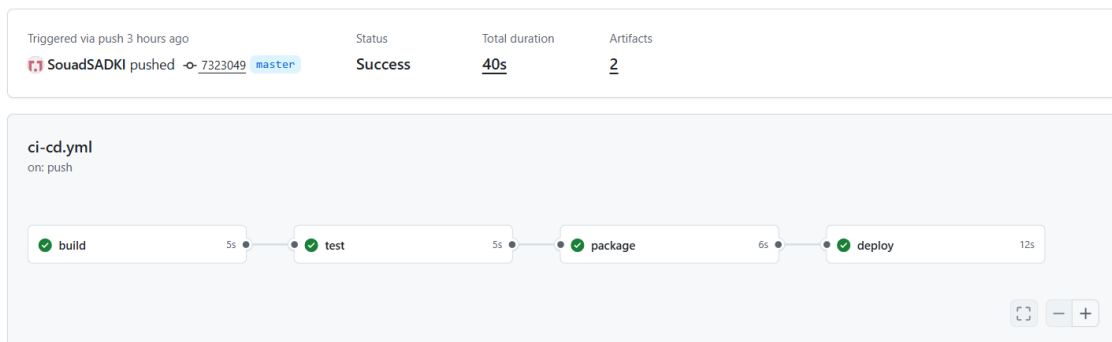
- Go to repository settings
- Navigate to Pages section
- Select GitHub Actions as deployment source
- Add deploy job in workflow using actions/configure-pages and deploy-pages



8. FINAL RESULT

After successful execution:

- CI pipeline validates code automatically
- CD pipeline deploys application
- Application becomes accessible via GitHub Pages URL



Test our pipeline :

1. Modify the HTML file

```
C: > Users > hp > Desktop > CI-CD BI-2SCL > <> index.html > html > head > title
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>My First Web App at ENSIAS |</title>
5  |   <link rel="stylesheet" href="style.css">
6  </head>
7  <body>
8  |
9  |   <h1>Simple CI/CD Demo</h1>
10 |
11 |   <button onclick="showMessage()">
12 |   |   Click Me
13 |   </button>
14 |
15 |   <p id="message"></p>
16 |
17 |   <script src="app.js"></script>
18 |
19 </body>
20 </html>
```

Push the new version of the HTML file :

```
hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git add .

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git commit -m "change4_html-ensias"
[master c5ca000] change4_html-ensias
1 file changed, 1 insertion(+), 1 deletion(-)

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 303 bytes | 303.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/SouadSADKI/ci-cd-2SCL_BI.git
7323049..c5ca000 master -> master

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ |
```

IS our pipeline going to detect this change :

The screenshot shows a GitHub Actions workflow run for the repository 'SouadsADKI / ci-cd-2SCL_BI'. The workflow is named 'Simple CI/CD Pipeline' and the specific run is 'change4_html-ensias #3'. The run status is 'Success' with a total duration of 45s and 2 artifacts. The workflow steps are: build (5s), test (5s), package (7s), and deploy (15s). The left sidebar shows the 'Summary' tab with links to 'All jobs', 'Run details', 'Usage', and 'Workflow file'. The top navigation bar includes links to 'Code', 'Issues', 'Pull requests', 'Actions', 'Projects', 'Wiki', 'Security and quality', 'Insights', and 'Settings'. The bottom of the image shows a browser address bar with the URL 'https://souadsadki.github.io/ci-cd-2SCL_BI/'.

← Simple CI/CD Pipeline
change4_html-ensias #3

Summary
All jobs
build
test
package
deploy
Run details
Usage
Workflow file

Triggered via push 1 minute ago
SouadsADKI pushed to c5ca000 master
Status: Success
Total duration: 45s
Artifacts: 2

ci-cd.yml
on: push

build (5s) → test (5s) → package (7s) → deploy (15s)

← ↻ https://souadsadki.github.io/ci-cd-2SCL_BI/

Simple CI/CD Demo

Click Me

Let's change the background of our html page :

```
C: > Users > hp > Desktop > CI-CD BI-2SCL > <> index.html > html > head
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>My First Web App at ENSIAS </title>
5      <link rel="stylesheet" href="style.css">
6  </head>
7  <body style="background-color: green;">
8
9      <h1>Simple CI/CD Demo</h1>
10
11     <button onclick="showMessage()">
12         Click Me
13     </button>
14
15     <p id="message"></p>
16
17     <script src="app.js"></script>
18
19 </body>
20 </html>
```

The results is :



CONCLUSION

This lab demonstrates a complete CI/CD workflow using GitHub Actions for a simple web application. It introduces automation principles used in modern DevOps pipelines.